📖 1. Series Overview

The **Linux Fundamentals** pathway on TryHackMe is an essential deep dive into the operating system that powers the majority of the world's servers and hacking tools. This series covers everything from basic navigation to advanced automation and package management.

📂 Part 1: The Basics of Navigation

In the first part, I learned how to move around the file system and interact with files without a mouse.

🛠️ Core Commands Mastered:

- `ls`: List files in the current directory.
- `cd`: Change directory (moving between folders).
- `cat`: Read the contents of a file directly in the terminal.
- `pwd`: "Print Working Directory" (where am I?).
- `find`: Search for specific files based on name or permissions.

Task 1:

Introduction: no answer needed

Task 2:

a) A Bit of Background on Linux

Research: What year was the first release of a Linux operating system?

Ans 1991

Task 3:

Interacting With Your First Linux Machine (In-Browser)

no answer needed

Task 4:

Running Your First few Commands

a) If we wanted to output the text “**TryHackMe**”, what would our command be?

echo TryHackMe

b) What is the username of who you’re logged in as on your deployed Linux machine?

Tryhackme

Task 5:

Interacting With the Filesystem!

a) On the Linux machine that you deploy, how many folders are there?

4

b) Which directory contains a file?

folder4

c) What is the contents of this file?

Hello World

d) Use the cd command to navigate to this file and find out the new current working directory. What is the path?

/home/tryhackme/folder4

Task 6:

Searching for Files

a) Use grep on “access.log” to find the flag that has a prefix of “THM”. What is the flag?

THM[redacted]

b) And I still haven’t found what I’m looking for!

No Answer Needed

Task 7:

An Introduction to Shell Operators

a) If we wanted to run a command in the background, what operator would we want to use?

&

b) If I wanted to replace the contents of a file named “passwords” with the word “password123”, what would my command be?

echo password123 > passwords

c) Now if I wanted to add “tryhackme” to this file named “passwords” but also keep “passwords123”, what would my command be

echo tryhackme >> passwords

d) Now use the deployed Linux machine to put these into practice

No Answer Needed

Task 8:

Conclusions & Summaries

No answer needed

Task 9:

a) Terminate the machine deployed in this room from task 3.

No answer needed

b) Join [Linux Fundamentals Part 2!]

No Answer Needed

⚙️ Part 2: System Interaction & Permissions

Part 2 focuses on how Linux manages users, permissions, and the hidden power of "piping" data.

```
root@ip-10-10-204-84:~# ssh tryhackme@10.10.31.147
The authenticity of host '10.10.31.147 (10.10.31.147)' can't be established.
ECDSA key fingerprint is SHA256:QMXh2XyvU7v3mto/vk3giXHsh0fKecN0WCLZ/T7uuO4.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '10.10.31.147' (ECDSA) to the list of known hosts.
tryhackme@10.10.31.147's password:
```

🔐 Understanding Permissions:

Linux uses a **Read (r), Write (w), Execute (x)** system for three types of users:

1. **Owner:** The person who created the file.
2. **Group:** A collection of users with shared access.
3. **Others:** Everyone else on the system.

🧪 Advanced Operators:

- **The Pipe (`|`):** Taking the output of one command and sending it to another (e.g., `cat file.txt | grep "password"`).
- **Redirection (`>` and `>>`):** Saving command output into a file instead of printing it to the screen.
- **`sudo`:** Executing commands with **Root** (administrator) privileges.

Task 3:

a). Explore the manual page of the ls command

I checked the manual page of “ls” command using “man ls” command.

No answer needed

b). What directional arrow key would we use to navigate down the manual page?

down

c). What flag would we use to display the output in a “human-readable” way?

{From the manual page I got the switch -h.}

-h

Task 4:

a). How would you create the file named “newnote”?

touch newnote

b). On the deployable machine, what is the file type of “unknown1” in “tryhackme’s” home directory?

By using the “file unknown1” command I got the file type as ASCII text.

ASCII text

c). How would we move the file “myfile” to the directory “myfolder”

From above, I learned that we can using any file using the “mv” command.

mv myfile myfolder/

d). What are the contents of this file?

Using the “cat” command I got the flag from the file.

A: THM[redacted]}

Task 5:

a). On the deployable machine, who is the owner of “important”?

To see the permission, owner and other details for every files within a directory we can use the “ls -la” command. Using this command I got user2 as the owner of the mentioned file.

user2

b). What would the command be to switch to the user “user2”?

For switching user we use the “su username” command.

su user2

c). Now switch to this user “user2” using the password “user2”

I run “su user2” command and then entered the password “user2” for switching to the user2.

No answer needed

d). Output the contents of “important”, what is the flag?

Using the “cat important” command I got the flag.

THM[redacted]

Task 6:

a). Read me!

No answer needed

b). What is the directory path that would we expect logs to be stored in?

From the article provided in this task, I have learned that all kinds of history are stored in the “/var/log” directory.

/var/log

c). What root directory is similar to how RAM on a computer works?

/tmp

d). Name the home directory of the root user

/root

e). Now apply your learning and navigate through these directories on the deployed Linux machine.

A: No answer needed

Task 7:

a). Proceed to the next task to continue your learning

No answer needed

Task 8:

a). Terminate the machine from task 2!

No answer needed

b). Join [Linux Fundamentals Part 3](#)!

No answer needed

🚀 Part 3: Automation, Packages, and Processes

The final part covers how to maintain a Linux system, install software, and manage background tasks.

📦 Package Management:

I learned how to use the **APT (Advanced Package Tool)** to install and update software:

- `sudo apt update`: Refresh the list of available software.
- `sudo apt install [package]`: Download and install new tools.

📋 Process Management:

- `ps`: View running processes.

- `top`: A real-time view of system resource usage (CPU/RAM).
- `kill`: Terminate a process that is frozen or unauthorized.

⏰ Automation with Cron:

I explored **Crontabs**, which allow users to schedule scripts or commands to run automatically at specific times (e.g., daily backups).

Task 3:

a). Create a file using Nano

Using the command “nano text_file_name” I created a text file.

A: No answer needed

b). Edit “task3” located in “tryhackme”’s home directory using Nano. What is the flag?

Using the command “nano task3” in the home directory I opened the file and got the flag.

THM[redacted]

Task 4:

a). Ensure you are connected to the deployed instance (machine_ip)

No answer needed

b). Now, use Python 3's "HTTPServer" module to start a web server in the home directory of the "tryhackme" user on the deployed instance.

I started the HTTPServer module using the command "python3 -m http.server".

No answer needed

c). Download the file http://machine_ip:8000/.flag.txt onto the TryHackMe AttackBox

I downloaded the file using the "wget http://machine_ip:8000/.flag.txt" command and then used "cat .flag.txt" to get the flag.

THM[redacted]

d). Create and download files to further apply your learning — see how you can read the documentation on Python3's "HTTPServer" module.

Use Ctrl + C to stop the Python3 HTTPServer module once you are finished.

No answer needed

Task 5:

a). Read me!

No answer needed

b). If we were to launch a process where the previous ID was “300”, what would the ID of this new process be?

In this task I have learned that the process ID are managed sequentially. So if the previous process ID is “X” then the next process ID will be “X+1”.

301

c). If we wanted to cleanly kill a process, what signal would we send it?

From the given article in this task I have learned that

- SIGTERM — Kill the process, but allow it to do some cleanup tasks beforehand
- SIGKILL — Kill the process — doesn’t do any cleanup after the fact
- SIGSTOP — Stop/suspend a process

SIGTERM

d). Locate the process that is running on the deployed instance (machine_ip). What flag is given?

Using the “ps -aux” command I got a lot of processes. That’s why later I used the “grep” command also to get the exact flag.

THM[redacted]

e). What command would we use to stop the service “myservice”?

From the provided article I have learned that to stop a service I should use “systemctl stop service_name” command.

systemctl stop myservice

f). What command would we use to start the same service on the boot-up of the system?

In this task I learned that if I want to start a service automatically whenever I restart my machine, I should run the command “systemctl enable service_name” and this should be disabled by the command “systemctl disable service_name”.

systemctl enable myservice

g). What command would we use to bring a previously backgrounded process back to the foreground?

We can use the “&” operator of a command to run the process in the background and then if we wanna see what is going there in that background process, we can use “fg” command to bring the process to the foreground in the terminal.

fg

Task 6:

a). Ensure you are connected to the deployed instance and look at the running crontabs.

No answer needed

b). When will the crontab on the deployed instance (machine_ip) run?

Using the “crontab -e” command I found a cron job “@reboot /var/opt/processes.sh” which indicates the cron process will start when the machine is rebooted.

@reboot

Task 7:

a). Since TryHackMe instances do not have an internet connection…this task only requires you to read through the material.

No answer needed

Task 8:

a). Look for the apache2 logs on the deployable Linux machine

I knew that the logs are kept in the directory “/var/log/”. That’s why I went to the directory and listed all the files and directories in that folder. There I got another folder named “apache2”. I discovered several log files.

No answer needed

b). What is the IP address of the user who visited the site?

Firstly I tried to read the first file which was “access.log” and found that I don’t have permission to read that file. Then I run the command “ls -la” to see the details of the files there. Then I found that “access.log.1” file can be accessed by the user “tryhackme”. I read that log file and found the required ip.

10.9.232.111

c). What file did they access?

From the log file I found that the ip user related to the ip “10.9.232.111” accessed the “/catsanddogs.jpg” file.

/catsanddogs.jpg

Task 9:

a). Terminate the machine deployed in this room from task 2.

No answer needed

🚩 5. Lab Evidence

Throughout these three modules, I completed several interactive challenges to prove my command-line proficiency.

📸 Terminal Proof

```
tryhackme@linux2:~$ touch newnote
tryhackme@linux2:~$ ls
important  myfile  myfolder  newnote  unknown1
tryhackme@linux2:~$
```

```
tryhackme@linux2:~$ file unknown1
unknown1: ASCII text
tryhackme@linux2:~$
```

```
tryhackme@linux2:~$ mv myfile myfolder
tryhackme@linux2:~$ ls myfolder/
myfile
tryhackme@linux2:~$
```

```
tryhackme@linux2:~$ cd myfolder && cat myfile
THM{FILESYSTEM}
tryhackme@linux2:~/myfolder$
```

```
tryhackme@linux2:~$ su user2
Password:
```

```
user2@linux2:/home/tryhackme$ ls
important  myfolder  newnote  unknown1
user2@linux2:/home/tryhackme$ cat important
THM{SU_USER2}
user2@linux2:/home/tryhackme$
```

```
user2@linux2:/$ cd var
user2@linux2:/var$ ls
backups  crash  local  log   opt  snap  tmp
cache    lib    lock   mail  run  spool  www
user2@linux2:/var$ cd log
user2@linux2:/var/log$ ls
alternatives.log    btmp.1                dmesg.3.gz  lastlog
alternatives.log.1  cloud-init-output.log dmesg.4.gz  private
amazon              cloud-init.log        dpkg.log    syslog
apache2             dist-upgrade          dpkg.log.1  syslog.1
apt                 dmesg                 journal     syslog.2.gz
auth.log            dmesg.0               kern.log    unattended-upgrades
auth.log.1          dmesg.1.gz            kern.log.1  wtmp
btmp                dmesg.2.gz            landscape
user2@linux2:/var/log$
```

```
user2@linux2:/$ ls
bin   dev  home  lib32  libx32      media  opt   root  sbin  srv  tmp  var
boot  etc  lib   lib64  lost+found  mnt    proc  run   snap  sys  usr
user2@linux2:/$ cd tmp/
user2@linux2:/tmp$ ls
snap.lxd
systemd-private-a097546fc9d945678a5cf921d7b92753-apache2.service-tChOug
systemd-private-a097546fc9d945678a5cf921d7b92753-systemd-logind.service-SWhIeh
systemd-private-a097546fc9d945678a5cf921d7b92753-systemd-resolved.service-yFSpzh
systemd-private-a097546fc9d945678a5cf921d7b92753-systemd-timesyncd.service-gvncJ
i
user2@linux2:/tmp$
```

```
user2@linux2:/$ ls
bin   dev  home  lib32  libx32      media  opt   root  sbin  srv  tmp  var
boot  etc  lib   lib64  lost+found  mnt    proc  run   snap  sys  usr
user2@linux2:/$ cd root
bash: cd: root: Permission denied
user2@linux2:/$
```

```
tryhackme@linux3:~$ wget 10.10.215.10:8000/.flag.txt
--2024-10-22 17:02:13--  http://10.10.215.10:8000/.flag.txt
Connecting to 10.10.215.10:8000... connected.
HTTP request sent, awaiting response... 200 OK
Length: 20 [text/plain]
Saving to: ‘.flag.txt.1’

.flag.txt.1                         100%[===================================================================>]      20  --.-KB/s    in 0.001s

2024-10-22 17:02:13 (16.1 KB/s) - ‘.flag.txt.1’ saved [20/20]

tryhackme@linux3:~$ cat .flag.txt
THM{WGET_WEBSERVER}
tryhackme@linux3:~$
```

```
tryhackme@linux3:~$ python3 -m http.server
Traceback (most recent call last):
  File "/usr/lib/python3.8/runpy.py", line 194, in _run_module_as_main
    return _run_code(code, main_globals, None,
  File "/usr/lib/python3.8/runpy.py", line 87, in _run_code
    exec(code, run_globals)
  File "/usr/lib/python3.8/http/server.py", line 1294, in <module>
    test(
  File "/usr/lib/python3.8/http/server.py", line 1249, in test
    with ServerClass(addr, HandlerClass) as httpd:
  File "/usr/lib/python3.8/socketserver.py", line 452, in __init__
    self.server_bind()
  File "/usr/lib/python3.8/http/server.py", line 1292, in server_bind
    return super().server_bind()
  File "/usr/lib/python3.8/http/server.py", line 138, in server_bind
    socketserver.TCPServer.server_bind(self)
  File "/usr/lib/python3.8/socketserver.py", line 466, in server_bind
    self.socket.bind(self.server_address)
OSError: [Errno 98] Address already in use
tryhackme@linux3:~$ wget http://10.10.215.10:8000/testfile.txt
```

🎓 6. Conclusion & Skills Gained

By completing this series, I have built a professional foundation in:

- Efficiently navigating the **Linux File System**.
- Managing **Users and Groups** via the terminal.
- Automating repetitive tasks using **Shell Scripts and Cron**.
- Securely managing **Root Access** and file permissions.